

File Tree Constructor

Course files manager

Introduction

Problem

- Organizing a course can be bothersome and tedious.

Solution

- Small CLI, capable of standardizing a file tree and a note format.

Config based approach

Example configuration file:

```
{  
  "encoding": "UTF-8",  
  "includeDateInFilename": true,  
  "includeClassNameInFilename": true,  
  "subdirs": ["class material", "course notes"],  
  "notesFileFormat": "md"  
}
```

Parse the config file

Configuration file retrieval and parsing:

```
public static Config getConfigFromFile(String filename) throws IOException {  
    try (Reader rd = new FileReader(filename, StandardCharsets.UTF_8);  
        BufferedReader brd = new BufferedReader(rd)) {  
        return mapper.readValue(brd, ch.heigvd.Config.class);  
    }  
}
```

Write config to file

Here's how the config is written back to a file:

```
public void writeConfigToFile(String filename) throws IOException {  
    try (Writer wr = new FileWriter(filename, StandardCharsets.UTF_8);  
        BufferedWriter bwr = new BufferedWriter(wr)) {  
        mapper.writeValue(bwr, this);  
    }  
}
```

Root command

```
public Config getConfig() {
    Config config;
    try {
        config = Config.getConfigFromFile(configFilename);
    } catch (java.io.IOException e) {
        System.out.println(
            "Failed to open Config file : No such file\nCreating a default config file");
        config = new Config();
        try {
            config.writeConfigToFile("config.json");
        } catch (java.io.IOException e1) {
            System.out.println(e1.getMessage());
        }
    }
    return config;
}
```

Create class

How to use

```
java -jar fileTreeConstructor.jar Create_class <name> <path>
```

Example

```
java -jar ./demo/fileTreeConstructor.jar Create_class DAI ./demo
```

How it works

```
Files.createDirectories(Path dir);
```

New course note

How to use

```
java -jar fileTreeConstructor.jar new_note [-c=<class_name>] <path> <title>
```

Example

```
java -jar ./demo/fileTreeConstructor.jar new_note -c DAI ./demo/DAI/course\ notes "example"
```


New course note

The path is insensitive to the presence or absence of a `/`.

```
if (path.charAt(path.length() - 1) != '/') path += "/";
```

Create copy of existing file

```
File tmpDir = new File(path + title_composed + "." + fileConfig.notesFileFormat);
if (tmpDir.exists()) {
    System.out.println("Error: File already exists at: " + tmpDir.getAbsolutePath());
    return 1;
}
```

Conlcusion